

AI-Enhanced Quality of Multimodal Physics Word Problems in Authentic Contexts Using the P-MAGIC App

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Abstract

This study aimed to develop an app called Physics Multimodal Authentic Generated Interactive Content Problem (P-MAGIC) to gather authentic contextual information by using object recognition and real-time sensors to help students learn the principles of centripetal acceleration in Physics. We implemented an instructional-based prompt generator that adjusts the difficulty level of problems in three levels, with the questions themselves being generated by the Generative Pre-trained version 4o (GPT-4o) model. Furthermore, to evaluate our app, we assessed the quality of the generated questions using automatic and human evaluations. The automatic evaluation demonstrated that the questions were relevant, easy to understand, and diverse. Additionally, feedback from teachers showed high ratings for relevance, completeness, and concept accuracy. Interestingly, problems incorporating both text and images, followed by text, images, and graphs, outperformed those that were based on text and graphs or text and tables. This highlights the positive impact of visual elements on conceptual understanding and student engagement. Furthermore, the strong correlation between automatic and

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human evaluations indicates that automated assessments can reliably reflect human judgement, making them a scalable tool for evaluations. By providing teachers with the tools to create personalized and context-rich problems in a variety of formats could potentially transform physics education.

Keywords

multimodal physics problems, generative AI, physics education, automatic question generation, authentic contexts

Introduction

In educational practice, teachers frequently employ questioning techniques to enhance students' comprehension. Question-driven instruction is essential, and the design of high-quality questions directly affects learning outcomes. Well-constructed questions can more effectively measure students' conceptual understanding, yet creating such questions is often a challenging and time-consuming task for teachers. With the advancement of artificial intelligence (AI), automatic question generation (AQG) systems have emerged as a promising solution to reduce teachers' workload and improve assessment quality (Shi et al., 2024). Before the development of large language models (LLMs), AQG research relied primarily on question-answer datasets such as SQuAD 2.0 and NQ, which provided context, question, and answer pairs for training (Zhang et al., 2021). Modern LLMs and GPUs have greatly expanded these capabilities, enabling AI systems to generate more complex and contextually meaningful problems.

Advances in graphics processing units (GPUs) and large language models (LLMs) have significantly improved AI performance in solving complex tasks. (Patel et al., 2021; Roy & Roth, 2015; Zhang et al., 2019). A rapidly growing area benefiting from these advancements is the automatic solving of math word problems (MWP) (Zhao et al., 2023). While hardware improvements allow AI to handle geometry and calculus with higher accuracy, gaps remain in question generation. For instance, Hwang and Utami (2024) integrated GPT with authentic contextual recognition to generate MWPs at various difficulty levels, but their work excluded multimodal elements. Despite these developments in math, automated physics question generation remains underexplored. A review of 71 studies found only one focused on physics-related questions, highlighting a significant research gap (Kurdi et al., 2020). Early attempts to bridge this gap include PhysWikiQuiz, which uses structured Wikidata to generate exam questions, and systems combining pattern matching with LLMs (Marchotridyo and Ruskanda, 2024; Scharpf et al., 2022). However, these systems primarily focus on formula applications or text-based items. Consequently, existing methods lack the authentic, real-world multimodal contexts—such as images, graphs, and sensor data—required to reflect how physicists actually analyze data.

Recent studies on AI-supported physics task generation further clarify the gap addressed in this study. Ghosh and Das (2023) proposed an automated pipeline that

converts physics word problems into algebraic representations using template prediction and named entity recognition, but their approach mainly targets symbolic transformation rather than pedagogically rich problem design. [El-Adawy et al. \(2024\)](#) extended this direction by presenting a teacher-oriented ChatGPT 3.5 workflow for physics problem generation, showing how conversational and style-aligned prompting can support instructors in creating new problems more efficiently. At the same time, [Küchemann et al. \(2023\)](#) found that while ChatGPT-supported task development by prospective physics teachers produced comparable correctness to textbook-based design, it showed lower clarity and weaker contextual embedding, which highlights the need for systems such as P-MAGIC that provide stronger pedagogical control, richer contextual grounding, and multimodal support.

To address these gaps, the present study introduces a new approach to automated physics question generation through the development of the P-MAGIC app. Unlike prior systems, P-MAGIC integrates real-world contexts through automatic object recognition and mobile sensor measurements (accelerometer and gyroscope), enabling the generation of physics problems grounded in authentic centripetal acceleration scenarios. In addition, the system produces multiple modalities—including text, images, graphs, and tables—offering a more authentic and contextually grounded learning environment compared with text-only question generators. This multimodal design represents a central contribution of our study and positions P-MAGIC as a more comprehensive solution for physics question generation. Because the design of P-MAGIC relies not only on automated generation but also on principles from physics education research, it is essential to situate this work within a strong theoretical foundation rooted in multimodality and social semiotics. Social semiotics explains how people create meaning when they interact with others. It focuses on the many ways a “text” can be produced, not only through written words but also through images, sound, and other forms. The term semiotics comes from the Greek word *sêmeion* (“sign”), where a sign is the smallest meaningful unit formed by linking a form (what we see or hear) with a meaning (the idea it represents). When signs are expressed through different modes, meaning becomes easier to notice and experience—something we can see, touch, or hear—and this allows people to interpret and engage with it through their senses ([Gualberto & Kress, 2025](#)).

Physicists routinely use diverse representations to conduct investigations and communicate findings, making representational fluency a fundamental component of physics practice. Therefore, students must learn to interpret and use multiple representations early in their education. A multimodal teaching approach—incorporating text, visuals, graphs, and tables—helps students construct meaning more effectively and sustain engagement ([Stefanel, 2019](#)). Learning physics requires more than memorizing facts; it involves understanding concepts, theories, terms, and methods, and applying them to investigate and explain physical phenomena. From this perspective, multimodal representations serve not only as communication tools but also as essential supports for conceptual understanding in physics education. See [Table 1](#) for a summary of related works on automatic question generation.

Table 1. Related Works of Automatic Question Generation

Reference	(Scaria et al., 2024)	(Wang et al., 2023)	(Ling & Afzaal, 2024)	(Polozov et al., 2015)	(Singhal et al., 2016)	(Marchotridyo and Ruskanda, 2024)
Topic	Data science course	Information security course	Program-ming I, big data, enterprise modelling course	Math formal domain	Physics and geome-try	Physics
Multimodali-ty	No	No	No	No	No	No
Annotation	No	No	No	No	No	No
Authentic context scenario	No	No	No	No	No	No
Difficulty level	Yes	No	No	No	Yes	Yes
Human type						
Human evaluation	Reviewed 9 criteria (understandable, topic relevancy, grammatical, clear, rephrase, answerable, central, would you use it, Bloom's level)	No	Reviewed 5 criteria (correctness, clarity, difficulty, and usefulness)	Crowdsourcing review 2 criteria (language comprehensibili-ty & mathematical applicability)	Evaluate the difficulty-ty level	A 1–3 scale was used: 1 for unsolvable, 2 for hard to understand, and 3 for clear and solvable questions
Automatic evaluation	Used gemini pro to automatically evaluate using the same nine criteria.	Automatic evaluation used BLEU and ROUGE-L	Used BLEU, METEOR, and ROUGE	No	No	BLEU, METEOR, and ROUGE, ParaScore
Num. of participants	2 teachers in data science and teaching	No	3 teachers in course topic	20 co-workers	10 teachers	1 researcher

Teaching science is fundamentally a multimodal communicative practice, where meaning is constructed through the coordinated use of multiple semiotic resources such as diagrams, graphs, equations, gestures, and written text (Jewitt, 2008; van Leeuwen, 2005). A social semiotic perspective emphasizes that each mode affords different possibilities for meaning-making and that scientific ideas become accessible only when learners interpret how these semiotic resources work together. Recent work in multimodal science education reinforces this view. Yeo and Nielsen (2020) argue that science instruction cannot rely solely on verbal or textual explanations because scientific concepts are inherently multimodal, requiring students to move across representations and understand how meaning is shaped by the mode used. Similarly, Wanselin, Danielsson, and Wikman (2022) demonstrate that multimodal texts in science require students to interpret how written language, symbolic notation, and visual elements interact—a process that can reveal students' conceptual understanding and guide formative assessment.

Building on this social semiotic foundation, we conceptualize Multimodal Physics Problems (MPP) as a form of instructional representation that intentionally combines text, images, graphs, tables, and mathematical expressions to support meaning-making in physics learning. In this study, we adopt MPP generation as a theoretical lens for exploring effective teaching strategies for centripetal acceleration in Grade 11. From this perspective, physics instruction should purposefully integrate diverse modes and support students in interpreting the relationships among them. The focus of this study is therefore to examine how MPP—constructed through coordinated multimodal representations—can enhance students' understanding of centripetal acceleration by making scientific meanings more explicit, interpretable, and aligned with real-world experience.

To assess the effectiveness of the MPP generation, we evaluate how well it generates questions and identify areas for improvement. This evaluation process also informs future app development as positive evaluation results can guide research directions. Our study aims to demonstrate that the proposed app performs well according to both automatic and human evaluation methods. Additionally, we investigate the consistency between these two evaluation approaches through Pearson correlation analysis. Based on this background, the study addresses the following research questions.

1. What are the results of the automatic evaluation of MPP generation?
2. What are the results of human evaluation of MPP generation?
3. Is there a significant correlation between the automatic evaluation scores and the human evaluation results of MPP generation?

Literature Review

Physics Word Problem

Physics word problems require not only numerical computation but also the correct application of physical principles. Such problems simultaneously examine learners'

grasp of physics concepts, their analytical reasoning, and their mathematical proficiency; success or failure can influence broader cognitive development (Abraham & Khemani, 2016). Typically, a physics word problem sets up a real-world situation and asks students to determine specific physical quantities from the information supplied.

Physics problems differ from standard math word problems (MWPs) in two major respects. First, they demand accurate recall and use of domain knowledge and units. Second, extra reasoning is often required before any calculation can begin (Ding et al., 2023); for example, consider a physics word problem that describes a child standing 1 m from the centre of a playground carousel that completes one full turn every 4 seconds. Although the question never gives the angular speed or centripetal acceleration directly. The student first has to figure out the spin speed (angular velocity) using $\omega = 2\pi \div T$, where T is the time for one full turn. Next, they use that spin speed to find the inward acceleration: $ac = \omega^2 \times r$. These two steps, working out ω from the period and then using $ac = v^2/r$ or $ac = \omega^2 r$, show the extra thinking often needed in physics questions about spinning motion. Because these steps are not stated directly, problems with centripetal acceleration and rotation are difficult. This makes them both challenging and useful for systems/apps that try to create and solve physics questions automatically. Both steps are challenging and useful for apps that try to create and solve physics questions automatically. Therefore, to help make these problems easier to understand and solve, our current app uses multimodal features (like images or diagrams).

The Importance of Various Representations/Multimodal in Learning and Understanding Physics

A multimodal representation refers to the combination of different types of data from multiple sources (Baltruaitis et al., 2019), and this definition provides an essential foundation for understanding how learning occurs in science. Teaching and learning naturally rely on multiple modes to communicate and make sense of ideas (Jewitt, 2012). Science education is therefore inherently multimodal because scientific meaning is expressed through a coordinated set of representations—mathematical symbols, diagrams, pictures, tables, graphs, and spoken or written language (Gilbert, 2010; Jewitt et al., 2001). From a pedagogical perspective, multimodal representations are not merely communication tools; they function as essential supports for conceptual understanding in physics. Airey (2009) argues that becoming competent in science means learning to access a discipline's "way of knowing," which is expressed through its representational system. Likewise, Lemke (2004) highlights that science is "naturally multimodal," meaning that no single representation can fully convey a scientific concept; instead, understanding emerges from navigating across multiple modes that each reveal different aspects of meaning. Within this framework, students must learn to interpret and extract information from various representations early in their physics education. Each mode foregrounds a distinct component of a concept—for example, graphs emphasize relationships, equations formalize quantitative structure, and images depict spatial or physical characteristics. By combining text, images, graphs, and tables

in our MPP, we aim to leverage these multimodal affordances to help students build deeper and clearer conceptual understanding of physics.

Contextualization and Personalization in MPP Generation

Our approach starts with a goal: let students tackle problems that feel real and familiar. Rather than abstract exercises, each question is rooted in everyday moments, the steady spin of an electric fan, a bicycle wheel rolling down the street, the washing machine or the outdoor fitness gear at a neighborhood playground (Uz Bilgin & Tokel, 2019). We weave these scenes into a continuous, interactive learning flow inside an app that guides students through real-world tasks. For example, before the app generates a set of MPPs, students are asked to measure the acceleration and angular motion of objects around their own school through our app. Because the task is hands-on and personally relevant, they immediately see why it matters. Personalization is the next layer. When the app tailors each MPP to the learner's own data and progress, students consistently outperform those who work with generic tasks. Results show that students perform better with personalized support than without personalized support. Tailoring the context boosts intrinsic motivation, so learners feel more autonomous, competent, and connected to the scenario (Fitzpatrick et al., 2019). Personalization of MPPs also involves generating problems at different difficulty levels so students with varying thinking skills and knowledge can work on them, so students with different levels of background knowledge can all find a comfortable starting point and room to grow.

Annotation

Annotation has long been recognized as an essential learning practice in science education because it enables students to engage actively with complex ideas. Through note-taking, marking, and adding interpretive comments, learners interact directly with learning materials in ways that strengthen understanding and memory (Nokelainen et al., 2003). This is particularly important in physics, where much of the content is abstract and mathematical. By annotating figures, graphs, diagrams, or equations, students externalize their thinking and make sense of concepts such as centripetal acceleration. Annotation also supports metacognitive awareness, allowing learners to monitor what they understand and where they need clarification (Hwang et al., 2007).

As the practice evolved, digital and collaborative annotation tools expanded the pedagogical potential of annotations. Systems like Annotea demonstrate how shared annotation environments enable learners to exchange comments, negotiate meaning, and co-construct knowledge while working on physics problems (Kahan & Koivunen, 2001). Such tools not only promote collaborative learning but also help students organize, revisit, and structure their notes in meaningful ways—an important benefit when dealing with physics' dense symbolic and representational content (Yang et al., 2004). These foundations illustrate why annotation is regarded as a powerful mechanism for promoting active engagement, representational fluency, deeper comprehension and boosts student motivation (Azouaou et al., 2004).

In this study, automated annotation is embedded directly into each modality produced by P-MAGIC. When a photo of a real-world rotating object is captured, the system automatically annotates key variables related to centripetal acceleration—such as radius, direction of centripetal acceleration, and angular velocity—so that learners can immediately identify the physical quantities represented in the scenario. Beyond images, the multimodal questions generated by P-MAGIC also include automated annotations within graphs, tables, and figures. For graphs and tables, the system highlights important phases of the data—such as increasing, stable, and decreasing trends—allowing learners to focus on critical features of the representation. Through these automated annotations, P-MAGIC helps students interpret multimodal information more effectively by making conceptual cues explicit across all representational forms.

Proposed System—P-MAGIC App

We developed our proposed system as a mobile application called Physics Multimodal Authentic Generated Interactive Content Problem (P-MAGIC) for Android devices. The system aims to automatically generate Multimodal Physics Problems (MPP) based on authentic contextual information. These physics problems come in three different difficulty levels – easy, intermediate, and advanced – to better support physics assessment in real-world learning situations.

Based on Figure 1, on the left top, the “Multimodality” bubble represents various input formats such as text, images, graphs, and tables, which align with the system’s capability to create physics problems in multiple formats. On the right top, the “Physics Word Problem” bubble signifies the final output, contextual physics problems generated through ChatGPT based on real-life input data. The arrow labelled “To Facilitator” suggests that the multimodal is a facilitator for word problems. The “Annotation” bubble connected to “To link” represents the automated annotation for each modality of the generated question. This annotation includes labelling or

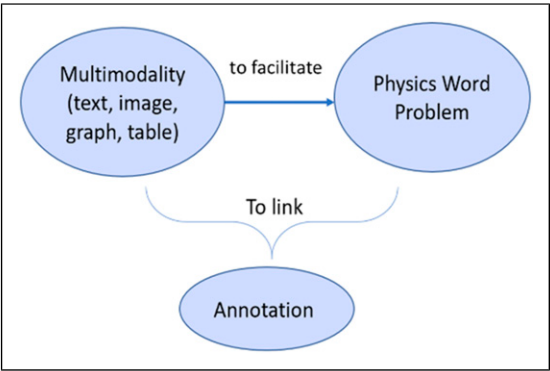


Figure 1. Multimodality physics problem framework

highlighting important variables, such as radius, angular velocity, and acceleration in images, as well as identifying different phases (e.g., increase, steady, decrease) in graph and table data. These annotations support multimodality by helping students better understand and interpret the content presented in the multimodal questions. The overall app is composed of three main modules. A. Integration of Real-World Sensor Data. P-MAGIC interfaces directly with mobile-device sensors, specifically the accelerometer and gyroscope, to capture real-time measurements from learners' interactions with physical objects. By grounding problem generation in authentic experimental sensor data rather than purely synthetic inputs, the system produces physics questions that are tightly coupled to the learner's actual motion context and measurement conditions. This capability addresses a fundamental limitation of standalone ChatGPT, which cannot natively access, synchronize, or validate live sensor streams without an external acquisition and processing layer. Consequently, real-world data integration is not an optional add-on but a core requirement for generating contextually meaningful physics problems that reflect genuine experimental variation and realism. B. Automated Multimodal Question Generation. Beyond text-based question construction, P-MAGIC automatically generates multimodal problem materials, including annotated figures, graphs, and tables, to support richer conceptualization and interpretation of motion phenomena. The system embeds automated annotations across modalities—for instance, indicating radius and direction vectors, highlighting angular velocity and centripetal acceleration cues, emphasizing graph patterns, and structuring tabular representations—so that learners can connect mathematical relationships to visual and empirical evidence. These outputs depend on dedicated pipelines for visualization, layout, and annotation that are not inherent to ChatGPT's default functionality. By operationalizing these multimodal components end-to-end, P-MAGIC enables physics problems that are more interpretable, instructional, and aligned with how experimental data are commonly analyzed in practice. C. Controlled Difficulty Levels as a personalization feature. To ensure instructional consistency and personalization, P-MAGIC produces problems at three structured difficulty levels (easy, intermediate, advanced) mapped to explicit learning objectives and Bloom's revised taxonomy. While ChatGPT can generate problems of varying complexity, it does not inherently guarantee stable difficulty control, objective alignment, or curriculum-consistent progression across generated items. P-MAGIC addresses this gap by encoding difficulty-level logic directly into its generation pipeline, enabling predictable scaling of cognitive demand and assessment focus. Taken together, these components demonstrate that P-MAGIC is not merely a wrapper around ChatGPT, but a purpose-built educational system that integrates authentic sensor-derived data, automated multimodal representations, systematic annotation, and pedagogically grounded difficulty control to generate physics problems with higher contextual fidelity and instructional coherence. [Figure 2](#) illustrates the system's overall workflow, while [Figure 3](#) details the specific steps in the MPP generation process.

Authentic Context Recognition and Information Acquisition

Our system provides functionalities to help learners obtain centripetal objects in their surrounding area and enable authentic contextual information acquisition. Our

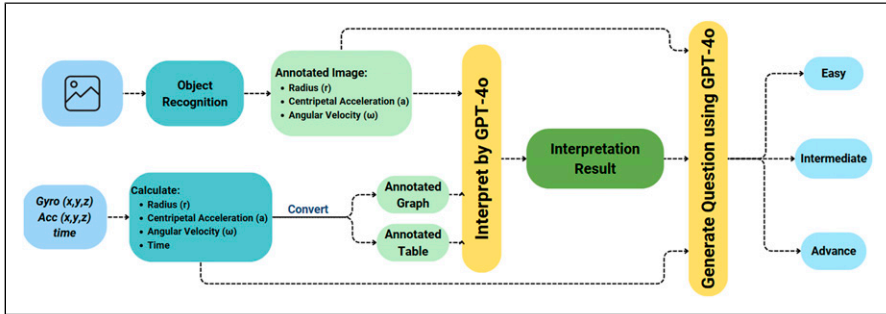


Figure 2. System development process

approach utilizes a camera function along with an object detection model provided by the ChatGPT Vision Model (4o) APIs to acquire contextual information. When an object is subsequently provided to the model, it will output a label of the object it detects. After that we also give automation annotations to every multimodality (image, graph, table). For automation circle annotation we used Hough transform and OpenCV to automatically annotate the variable names (radius, centripetal acceleration, angular velocity) inside the object image. For automation annotation (graph & table) done by

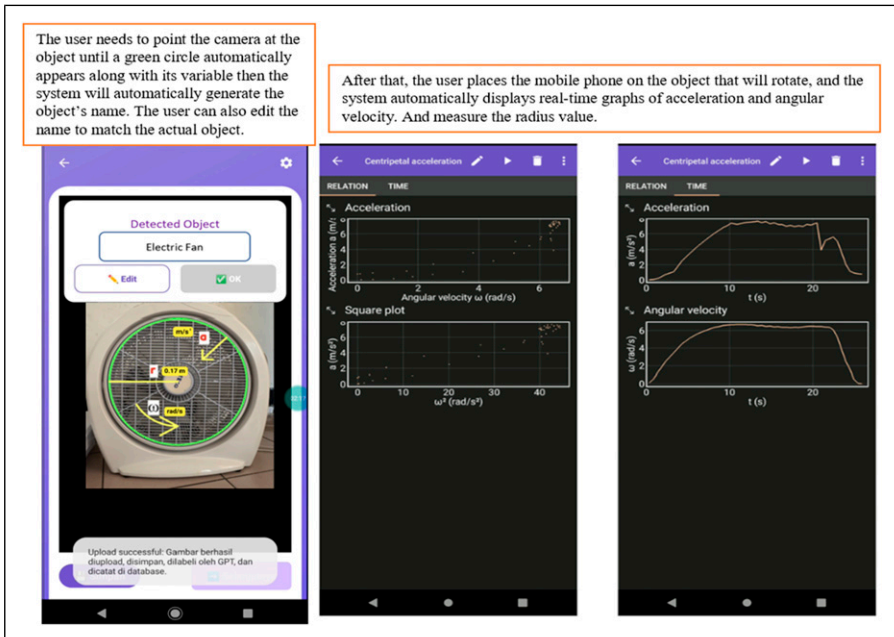


Figure 3. Authentic context recognition and mobile application interface that visualizes data through real-time sensor

calculating simple statistics (standard deviation and slope) in moving windows and then labelling phases by comparing these to thresholds. This is a threshold-based phase detection method. Besides, we also employ the Android sensor (gyroscope sensor and accelerometer sensor) by Phypox (Staacks et al., 2018) to get the real-time data of centripetal acceleration and angular velocity.

Instructional-Based Prompt Generation with Different Difficulty Levels

In this study, we used ChatGPT powered by the GPT-4o language model, which contains over 200 billion parameters, to generate physics word problems focused on centripetal acceleration. Instead of retraining or fine-tuning the model, we guided it through well-crafted natural language prompts and examples to help it understand the task. These prompts acted as instructions, showing the model how to create questions that resemble those written by humans (Moore et al., 2022). To support the generation of relevant problems, we built a system that creates instructional prompts based on the user's input and the context provided. Prior to finalizing the prompt templates, we experimented with different versions of the prompts to see which ones led to the most accurate and meaningful outputs. Our system covers a wide range of centripetal acceleration problems, from easy to advanced levels. It includes contexts like rotating playground equipment, electric fans, vegetable washers, rotating displays, and bicycle wheels. The difficulty levels follow the Indonesian Ministry of Education's high school physics curriculum, which supports strong understanding of centripetal motion. At the easy level, problems focus on remembering and understanding key concepts like angular velocity, radius, and acceleration. The intermediate level requires students to apply these concepts and perform calculations. Advanced problems involve comparing two scenarios, analyzing the differences, and solving multi-step reasoning tasks. This approach uses Bloom's revised taxonomy (Anderson & Krathwohl, 2001) to ensure the problems match cognitive skill levels: easy (remembering, understanding), intermediate (applying), and advanced (analyzing) comparison scenarios. The interface of MPPs generation for three difficulty levels is shown in Figure 4.

Question Generation by GPT

We developed MPP generation system across four main components: the front end, server side, back end, and database. The user interface was created as an Android application, allowing users to interact directly with the system. The front end includes all visible features and functionality, built using Java for logic and XML for designing the user interface. To handle client requests from the app, we implemented a RESTful API that acts as a bridge between the front end and the server. The API enables smooth communication and makes the system modular and scalable. On the server side, we integrated the ChatGPT API powered by the GPT-4o model and deployed it using a Virtual Private Server (VPS). This setup allowed us to access AI-generated responses for MPP generation via internet-based API calls. Since ChatGPT is highly responsive to

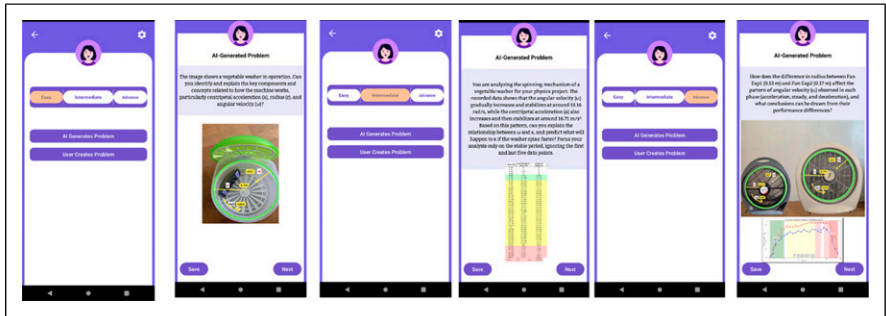


Figure 4. MPPs generation for three difficulty levels (easy, intermediate, and advanced)

how input prompts are phrased, we carefully crafted clear and unambiguous instructional prompts to guide the model. We went through multiple iterations to refine these prompts, aiming to improve the content quality of the generated questions. Finally, we stored the generated word problems in a MySQL database for future retrieval and use. An example of the system’s output is illustrated in a block diagram of MPP generation in Figure 5. The block diagram outlines the three main functional modules of the P-MAGIC system.

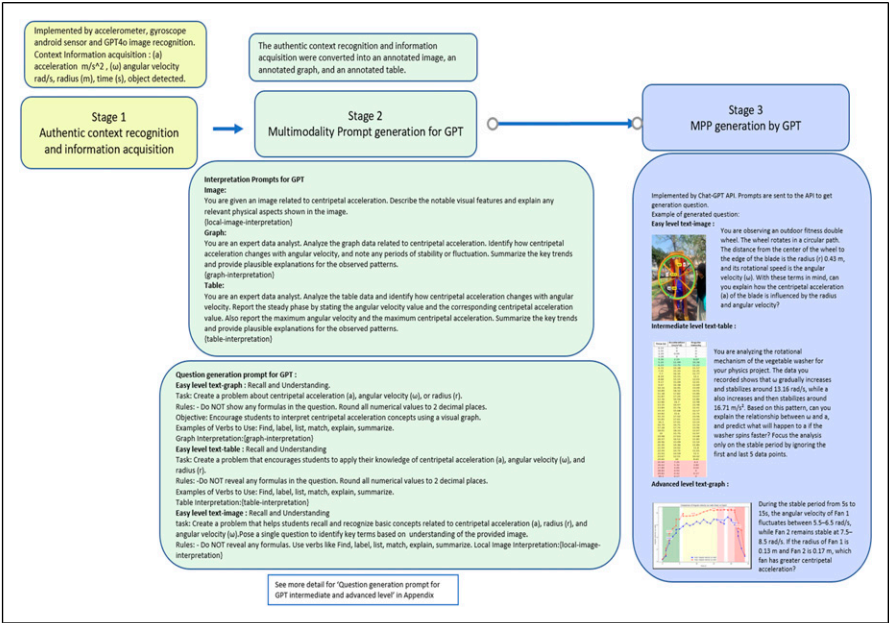


Figure 5. Block diagram of MPP generation (for more detail at intermediate and advanced levels, see The Appendix)

1. **Authentic Context Recognition and Information Acquisition:** This initial module gathers real-life input data, such as identifying centripetal objects via object detection using the ChatGPT Vision Model (4o) APIs and collecting real-time data like centripetal acceleration and angular velocity using Android sensors (gyroscope and accelerometer).
2. **Instructional-Based Prompt Creation:** This module transforms the collected real-life contextual information into clear, structured instructional prompts specifically for ChatGPT. This module is critical for managing question complexity, as it includes three separate prompt generators designed for easy, intermediate, and advanced difficulty levels.
3. **MPP Generation:** Utilizing these structured prompts, this final module generates the Multimodal Physics Problems (MPP) in various formats, including text, images, graphs, and tables, making the questions both accessible and engaging for students.

Methodology

This study employed a dual evaluation framework combining automatic semantic metrics and expert human judgment to assess the quality of the Multimodal Physics Problems (MPPs) generated by the P-MAGIC system. The following subsections describe the participants, procedure, human evaluation, and automatic evaluation methods.

Participants

Ten senior high school physics teachers from different schools were recruited to evaluate the quality of the generated MPPs. All teachers had more than four years of teaching experience and were routinely responsible for designing written assessments aligned with national curriculum standards. Although they had not received specialized training in multimodal question design, all had prior preparation during their teacher education programs in constructing physics problems based on curricular learning objectives and conceptual learning goals.

In their regular teaching practice, these teachers frequently incorporate diagrams, graphs, and other representational forms when assessing topics such as motion, forces, and centripetal acceleration. Their professional background positions them as qualified evaluators for judging the linguistic, conceptual, and representational quality of the generated MPPs. All participants were informed of the study purpose, evaluation procedures, and confidentiality assurances prior to participation, and provided informed consent.

Procedure

A total of 150 MPPs were used in this study. All content was translated into Bahasa Indonesia to ensure accessibility without altering semantic intent. Teachers evaluated

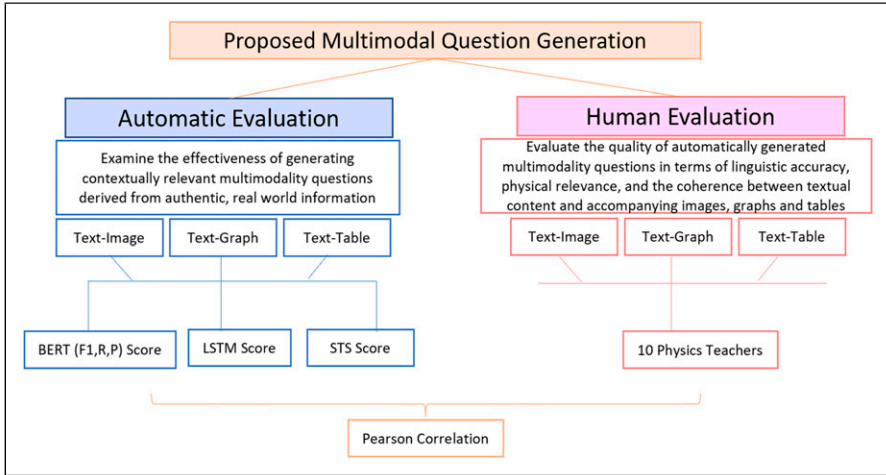


Figure 6. The evaluation framework

each MPP individually using a structured scoring rubric. Figure 6 presents the evaluation framework used in the study. Two complementary methods were applied.

- Automatic evaluation, using semantic similarity metrics
- Human evaluation, using expert teacher ratings

This combined approach enables a more robust assessment of question quality by integrating computational measures with pedagogically grounded human judgment.

Metrics for Automatic Evaluation

Our automatic evaluation used three metrics, BERT, STS, and LSTM as below.

BERT Score

We evaluated the capability of the system in terms of generating relevant questions from authentic contextual information using the BERT metric proposed by Devlin et al. (2019). The rise of contextual language models such as BERT represents a major shift in Natural Language Processing (NLP). Earlier evaluation metrics relied on surface-level word matching, which struggled to capture meaning when different words expressed the same idea. With the introduction of contextual embeddings in the BERT model (Devlin et al., 2019), NLP gained the ability to represent words based on surrounding context rather than in isolation. This development established a theoretical foundation

for evaluating semantic similarity in generated text, making BERT-based metrics highly influential within modern language generation research.

BERTScore (Zhang et al., 2020) builds directly on this theoretical advancement by using contextual embeddings to evaluate similarity between generated text and reference text. Instead of matching n-grams like BLEU, BERTScore compares semantic meaning at the token level. It works as follows: both candidate and reference texts are tokenized using the BERT tokenizer, embedded using a pre-trained contextual model, and then compared via cosine similarity to construct a similarity matrix. Greedy matching is used to compute precision (matching candidate tokens to reference tokens) and recall (matching reference tokens to candidate tokens), followed by an F1 score that combines both measures. Because it evaluates meaning rather than exact word overlap, BERTScore is more robust for assessing question generation systems and better reflects the semantic equivalence of paraphrased answers or varied expressions.

STS

Semantic Text Similarity (STS) originates from vector space models used to quantify how semantically related two pieces of text are. Its significance lies in providing a numerical measure of how closely two sentences align in meaning, independent of their surface structure. In educational question generation, STS is particularly useful for analyzing both accuracy (alignment with an instructional prompt) and diversity (variation between generated questions) (Mazidi & Nielsen, 2014). In our study, STS is used to compute semantic relatedness between instructional prompts and the generated multimodal physics problems, as well as the average semantic similarity across generated questions. A lower average similarity score indicates greater linguistic diversity. STS values range from 0 to 1, where 0 represents no shared meaning and 1 represents complete semantic equivalence, with scores such as 0.6 indicating moderate similarity and 0.8 indicating strong similarity.

LSTM

Long Short-Term Memory (LSTM) networks originate from the family of Recurrent Neural Networks (RNNs), which were developed to handle sequential data such as text. Traditional RNNs struggled with long sequences due to vanishing and exploding gradient problems. LSTM addressed these limitations through memory cells capable of selectively retaining important information and forgetting irrelevant input over long spans (Hochreiter & Schmidhuber, 1997). This innovation made LSTM a foundational model in early neural language processing and a precursor to later transformer-based methods. In this study, LSTM is used as an encoder to capture semantic information from both reference texts and system-generated questions. Following Luo et al. (2021), LSTM extracts meaningful features from sequences after tokenization and padding. Using TensorFlow and Keras, the network processes the sequences through an embedding layer, an LSTM layer, and dense layers to produce fixed-size vector representations. Cosine similarity between these embeddings provides a measure of semantic

Table 2. Human Evaluation Criteria

Dimension	Criteria	Reference	Description
Linguistic	Relevancy	(Kurdi et al., 2020)	Assessed whether the question contains relevant keywords, named entities, and question types to evaluate its relevance
	Linguistic complexity	(Steuer et al., 2021)	Determine whether the generated question is easy to understand.
	Diversity	(Kurdi et al., 2020)	Diversity refers to the range of vocabulary, sentence structures, and questions used in the generated question.
	Completeness	(Kurdi et al., 2020)	We measure the question's inclusion of all necessary components, such as context, conditions, and sufficient detail.
	Fluency	(Gatt & Krahmer, 2018)	The assessment includes checking whether the generated question is grammatically correct
Physic aspect	Answerability	(Scaria et al., 2024)	Assesses whether the question provides all the necessary information or context for students to solve it and whether it can be answered using valid physics principles.
	Difficulty level	(Venkatesh et al., 2022)	Measures the cognitive challenge of the question, whether the question is already at the appropriate level (easy, intermediate, advanced)
	Realistic	(Venkatesh et al., 2022)	Assesses whether the question setup or scenario is plausible and physically consistent with real-world situations.
	Concept understanding	(Horbach et al., 2020)	This assesses how well the question tests, reflects, or integrates physics concepts.
	Variable's name consistency	(Marchotridyo and Ruskanda, 2024)	Evaluates whether variable names (e.g., v for velocity, a for acceleration) are logically and consistently used, following physics notation standards.

(continued)

Table 2. (continued)

Dimension	Criteria	Reference	Description
Image - text question	Precision	(Otani et al., 2023)	Assesses how accurately and clearly the question refers to the image.
	Relevancy	(Otani et al., 2023)	Assesses how well the image supports or aligns with the question’s content, concept, or objective.
Text - graph question	Graph accuracy and proper labeling	(Stefanel, 2019)	Assesses whether the graph correctly plots all data points and accurately reflects the intended physical phenomena. Additionally, it assesses whether the axes clearly label the correct physical quantities and whether the units are present and consistent with the data.
	Scale and proportions	(Stefanel, 2019)	Assesses whether the graph uses an appropriate scale with evenly spaced intervals that match the data range, ensuring readability.
	Graph sense in physical contexts	(Stefanel, 2019)	Measures whether the conclusions or answers derived from the graph are consistent with known physical principles or laws. Determines whether the graph logically aligns with the physical scenario.
	Creativity and critical thinking	(Stefanel, 2019)	Assesses whether the question encourages deeper insights based on the interpretation of graphical data.
	Relevancy (axis tittle, value in question text-graph relate)	(Stefanel, 2019)	The system evaluates the degree to which the values and axis titles in the graph align with the values and context specified in the question text.

(continued)

Table 2. (continued)

Dimension	Criteria	Reference	Description
Text- table question	Clear labels and scales	(Stefanel, 2019)	Measured if each row and column is clearly labelled with corresponding physical quantities and that units of measurement are present and accurate. And whether the data in the table is presented using an appropriate scale for accurate reading.
	Proportional and predictive reasoning	(Stefanel, 2019)	The table's ability to facilitate reasoning about proportional relationships between variables and comprehend the impact of changes in one quantity on another is crucial for comprehending numerous physics concepts. And whether students are encouraged to make logical predictions from the data
	Connection to physics concepts and principles	(Stefanel, 2019)	The assessment evaluates the table's foundation in fundamental physics concepts and its direct connection to principles.
	Real-world applicability and context	(Stefanel, 2019)	It assesses the effectiveness of the table in reflecting real-life physics situations, thereby enhancing students' ability to connect theory to practice using authentic data.
	Relevancy	(Stefanel, 2019)	Measures how well the table's column titles and values align with and support the context and content presented in the accompanying question text.

alignment, enabling us to evaluate how closely the generated questions match reference questions not only lexically but also contextually.

Human Evaluation

Our human evaluation was conducted using a 22-item rubric based on prior research in question generation, physics instruction, and multimodal learning (see [Table 2](#). Each

criterion was rated on a 10-point Likert scale, examining the quality of the MPPs across five main dimensions.

1. Linguistic Quality
(Relevance, complexity, diversity, completeness, fluency).
2. Physics Dimension
(Answerability, difficulty level, realistic, concept understanding, variable consistency).
3. Image–Text question
(Precision and relevance of imagery relative to question text).
4. Graph–Text Alignment
(Graph accuracy, labeling, scale, physical consistency, creativity and critical thinking, and relevancy).
5. Table–Text Alignment
(Label clarity, scaling, proportional and predictive reasoning, physics conceptual connection, real-world applicability, relevancy).

Teachers scored each MPP independently. To examine inter-rater consistency, Cohen’s kappa (McHugh, 2012) was used as a measure of inter-rater reliability.

To maintain Pedagogical Integrity, our study directly addresses Item 25 (AI Risks) and Item 28 (Unintended Consequences) of the RAISE framework (Allison, 2026). By combining instructional-based prompt engineering with human-in-the-loop validation, we ensured high concept accuracy ($\kappa = 0.89$). These safeguards mitigate AI hallucinations and prevent the loss of learner agency, ensuring the P-MAGIC app remains a trustworthy educational tool.

Result and Discussion

Automatic Evaluation

In the automatic evaluation presented in Table 3, we assessed the quality of 150 system-generated MPPs using five key metrics: BERT F1, BERT Recall, BERT Precision

Table 3. Automatic Evaluation Results for Different Difficulty Levels of MPPs

Difficulty level	n	Statistic	BERT F1 score	BERT R	BERT P	Diversity score	LSTM score
Easy	60	M	0.877	0.881	0.875	0.503	0.814
		SD	0.018	0.037	0.027	0.073	0.083
Intermediate	55	M	0.885	0.889	0.882	0.52	0.835
		SD	0.015	0.036	0.023	0.051	0.087
Advanced	35	M	0.883	0.876	0.892	0.508	0.823
		SD	0.015	0.038	0.0306	0.045	0.087
Total	150	M	0.882	0.883	0.882	0.514	0.824
		SD	0.015	0.038	0.0306	0.045	0.087

(semantic similarity measures), Diversity Score, and LSTM Score (language fluency/quality). These metrics offer a holistic view of both the semantic alignment and linguistic quality of the generated problems across three difficulty levels: easy, intermediate, and advanced.

The overall mean BERT F1 score of 0.882 indicates strong semantic relevance between the generated MPPs and their instructional prompts. This is further supported by consistent BERT Recall (0.883) and BERT Precision (0.882) values, showing a balance between completeness and exactness in semantic content. Across difficulty levels, the BERT scores remain high and stable (easy: 0.877, intermediate: 0.885, advanced: 0.883), reflecting the system's ability to generate conceptually aligned and meaningful physics problems at all levels.

The diversity score averaged at 0.514, which is higher than conventional text generation systems and suggests notable variation in sentence structure and content across the generated MPPs. This diversity ensures that students are exposed to a wide range of problem formulations, an essential aspect in educational contexts for fostering critical thinking and adaptive reasoning (Zhang et al., 2019). The LSTM Score, which assesses sentence fluency and grammatical coherence, averaged 0.824, further indicating that the questions are not only semantically accurate but also linguistically fluent and natural-sounding. We observed the highest fluency at the intermediate level (0.835), followed by advanced (0.823) and easy (0.814), with a marginal variation.

Overall, these results show that the MPP generation system can successfully create high-quality, varied, and relevant questions that are well-organized and suitable for different difficulty levels. While automatic metrics offer strong indications of performance, human evaluations are still recommended to validate the pedagogical appropriateness and clarity of the questions (Luo et al., 2021). Even though the conditions for parametric testing were satisfied, as shown by Levene's Test indicating similar variances ($p > 0.05$), the one-way ANOVA showed no significant differences in average scores among the six groups ($p = 0.455 > 0.05$). Even so, we continued with a Kruskal-Wallis test to double-check, knowing it can detect differences in medians or shapes of data that ANOVA might miss. The Kruskal-Wallis test yielded a statistically significant result ($H = 22.528$, $p < 0.001$), indicating meaningful differences in rank distributions among the groups. This result justified further post-hoc pairwise comparisons using the Mann-Whitney U test (Table 4). Out of all comparisons, two pairs demonstrated significant differences at the $p < .05$ level. First, Text_Image significantly outperformed Text_Graph ($p = .001$), and second, Text_Image_Graph showed higher mean ranks than Text_Table ($p = .018$). These findings highlight that the presence of images, either alone or combined with graphs, substantially enhances performance compared to text with graphs or tables. This supports the idea that using visuals, especially images, is very effective for improving understanding in physics problems that use different formats, which matches research showing that images from real life can help people think better and stay more engaged.

To look at differences between groups in six ablation annotation conditions based on relevancy scores, we first performed Levene's Test for Homogeneity of Variance, which showed a significant result ($p = .008$ based on mean), meaning the assumption of equal variances was not met. Therefore, we could not use standard one-way ANOVA and

Table 4. Automatic Evaluation Results for Different Ablation Modalities of MPPs

Pairwise comparison	Z	p-value	Higher mean rank
Text_Graph vs Text_Image	−3.269	0.001**	Text_Image
Text_Graph vs Text_Image_Graph	−1.445	0.148	Text_Image_Graph
Text_Graph vs Text_Image_Table	−1.014	0.311	Text_Graph
Text_Graph vs Text_Table	−1.377	0.168	Text_Graph
Text_Graph vs Text_Table_Graph	−0.798	0.425	Text_Table_Graph
Text_Image vs Text_Image_Graph	−0.625	0.532	Text_Image
Text_Image_Graph vs Text_Image_Table	−1.827	0.068	Text_Image_Graph
Text_Image_Graph vs Text_Table	−2.358	0.018*	Text_Image_Graph
Text_Image_Graph vs Text_Table_Graph	−0.677	0.498	Text_Image_Graph
Text_Image_Table vs Text_Table	−0.128	0.898	Text_Image_Table
Text_Image_Table vs Text_Table_Graph	−1.429	0.153	Text_Table_Graph
Text_Table vs Text_Table_Graph	−1.759	0.079	Text_Table_Graph

instead used more reliable methods, Welch’s ANOVA and the Brown-Forsythe test. Both tests showed significant differences between the groups (Welch’s: $F(5, 106.76) = 19.49$, $p < .001$; Brown-Forsythe: $F(5, 180.44) = 17.82$, $p < .001$), indicating that at least one condition had a different mean score. Therefore, we deemed standard one-way ANOVA inappropriate and applied robust alternatives such as Welch’s ANOVA and the Brown-Forsythe test. Both tests revealed significant differences across groups (Welch’s: $F(5, 106.76) = 19.49$, $p < .001$; Brown-Forsythe: $F(5, 180.44) = 17.82$, $p < .001$), suggesting that at least one condition differed in mean score. Descriptive statistics showed that the highest mean relevancy score was achieved by the image with variable and value annotations condition ($M = 0.896$), followed by the table with highlight annotations ($M = 0.871$) and the graph with highlight annotations ($M = 0.860$). The lowest score was recorded under the Table without annotation condition ($M = 0.822$), indicating that lack of annotation led to lower semantic alignment (Table 5).

We performed Games-Howell post hoc tests (Table 6) to further investigate the differences due to unequal variances. The results confirmed that the image with variable

Table 5. Descriptive Statistic for Ablation Annotation

Condition	N	M	SD
Graph with annotation	40	0.859	0.052
Graph without annotation	40	0.825	0.042
Image with annotation	40	0.895	0.044
Image without annotation	40	0.839	0.022
Table with annotation	40	0.871	0.029
Table without annotation	40	0.822	0.058

Table 6. Automatic Evaluation Result for Ablation Annotation

Ablation		MD	p-value.	Cohen's d
Graph with annotation (highlight different time phases & text labelling annotation)	Graph without annotation	0.035*	0.018	0.723
	Image with annotations	−0.036*	0.016	−0.75
	Image without annotations	0.021	0.209	0.541
	Table with annotation	−0.012	0.829	−0.293
	Table without annotation	0.037*	0.038	0.673
Graph without annotation	Image with annotations	−0.071***	<0.001	−1.628
	Image without annotations	−0.014	0.410	−0.438
	Table with annotation	−0.046***	<0.001	−1.278
	Table without annotation	0.002	1.000	0.06
Image with annotation (variable and value annotation)	Table with annotation	0.024*	0.048	0.649
	Table without annotation	0.073***	<0.001	1.431
	Image without annotation	0.056***	<0.001	1.7
Image without annotations	Table with annotation	−0.032***	<0.001	−0.032
	Table without annotation	0.017	0.538	0.017
Table with annotation (highlight different time phases & text labelling annotation)	Table without annotation	0.049***	<0.001	1.13

*The mean difference is significant at the 0.05 level. **, Correlation is significant at the 0.01 level.

***The mean difference is significant at the 0.001 level.

and value annotation conditions performed significantly better than all other conditions ($p < .05$). Additionally, both Table with highlight annotation and Graph with highlight annotation were significantly superior to their non-annotated counterparts (Table: MD = 0.049, $p < .001$; Graph: MD = 0.035, $p = .018$). While homogeneity of variance was violated, the use of robust statistical methods provided strong evidence that explicit annotations including variables, values, and highlights substantially enhance the semantic relevance of multimodal physics problem presentations. These findings emphasize the importance of visual labelling in improving the clarity and effectiveness of content question generation.

Human Evaluation

As we mentioned earlier, the ten teachers assessed the quality of MPPs using a 25-item rubric with a 10-point Likert scale (see Table 2). To ensure consistent evaluation among the raters, inter-rater reliability was calculated using Cohen's Kappa, resulting in an average κ value of 0.76. According to (McHugh, 2012), this value indicates a good level of agreement. Table 7 displays the results of human evaluation. Human evaluators rated most of the MPPs created by the system as relevant. The average relevancy scores were 8.58 for the easy level, 8.21 for the intermediate level, and 8.18 for the advanced level. This means that the system is effective at creating MPPs that match the context. In simple terms, the prompt generator we used helps the GPT-4o model understand real-life situations and create relevant MPPs based on that information.

When looking at the language complexity, Table 7 shows that the easy-level MPPs got a lower average score (1.54) than the intermediate-level (2.47) and advanced-level (4.41) MPPs. This means that the easy-level problems are written more simply and are easier to understand compared to the others. This is important because our goal is to make problems with different difficulty levels to match students' learning needs. Our system can be useful not just for testing but also for supporting physics learning and studying how students understand different types of problems. Having questions at different difficulty levels allows for more personalized teaching, which helps both teachers and students.

Most of the questions received clear ratings for completeness, providing sufficient information to solve the problems. The average scores for completeness were 7.67 for easy, 8.81 for intermediate, and 8.54 for advanced problems. Completeness is important because it shows whether the question includes all the details needed to understand it. These results show that our system can make MPPs with enough context and clear meaning.

The average score for diversity was 5.42 (easy), 5.54 (intermediate), and 5.51 (advanced). These show that the questions were not too similar to each other. A higher score closer to 10 would mean that many questions are almost the same. For language fluency, the average scores were 8.60 for easy, 8.12 for intermediate, and 8.20 for advanced levels. This means most MPPs were written clearly, were easy to understand, and had no major grammar mistakes. In addition, Table 7 shows how the questions perform on five physics-related points. Answerability scores are highest for the easy set ($M = 8.69$, $SD = 0.47$) and fall for the intermediate ($M = 7.85$) and advanced tasks ($M = 7.28$). We expect this drop because harder problems require more steps, making some items feel less directly solvable. For difficulty level, the trend moves the other way; teachers rate easy questions at 5.40, intermediate at 6.35, and advanced at 7.43 ($\kappa = 0.75$), confirming that the system successfully grades the tasks from simple to demanding. Realistic context receives very high marks across the board (all means ≈ 8.20 ; $\kappa = 0.96$), showing the situations match everyday physics well, no matter the level. Concept understanding also stays strong (overall $\kappa = 0.89$); even the advanced set averages 8.18, indicating that the core physics ideas are still clear. Finally, variable-name consistency holds steady ($\kappa = 0.87$; means ≈ 8.60 for easy and ≈ 8.18 for higher

Table 7. Human Evaluation Results

Dimension	Sub-dimension	Cohen's Kappa				Easy level				Intermediate level				Advanced level			
		N		M		SD		N		M		SD		N		M	
Linguistic	Relevancy	0.890	60	8.585	0.555	55	8.211	0.925	35	8.180	0.925						
	Complexity	0.732	60	1.541	0.236	55	2.476	0.242	35	4.417	0.244						
	Diversity	0.877	60	5.417	0.288	55	5.539	0.102	35	5.511	0.133						
	Completeness	0.682	60	7.667	0.150	55	8.811	0.443	35	8.543	0.852						
	Fluency	0.967	60	8.603	0.528	55	8.125	0.906	35	8.200	0.933						
Physic aspect	Answerability	0.702	60	8.687	0.474	55	7.851	0.215	35	7.277	0.519						
	Difficulty level	0.750	60	5.404	0.167	55	6.353	0.209	35	7.431	0.237						
	Realistic	0.961	60	8.598	0.530	55	8.200	0.951	35	8.200	0.933						
	Concept understanding	0.890	60	8.585	0.555	55	8.211	0.925	35	8.180	0.925						
	Variable's name consistency	0.868	60	8.602	0.549	55	8.211	0.925	35	8.180	0.925						
Image - text question	Precision	0.663	15	8.200	0.409	12	8.725	0.403	7	8.986	0.038						
	Relevancy	0.830	15	8.173	0.440	12	8.708	0.429	7	8.986	0.038						
Text - graph question	Graph accuracy & proper labeling	0.650	17	8.035	0.337	17	8.872	0.267	9	8.989	0.033						
	Scale and proportions	0.718	17	8.247	0.352	17	8.872	0.267	9	8.989	0.033						
	Graph sense in physical context	0.725	17	8.253	0.362	17	8.872	0.267	9	8.989	0.033						
	Creativity and critical thinking	0.744	17	8.265	0.386	17	8.872	0.267	9	8.978	0.067						
	Relevancy	0.817	17	8.129	0.396	17	8.856	0.315	9	8.978	0.067						
Text- table question	Clear labels scale	0.797	11	8.208	0.405	11	8.000	0.000	7	8.943	0.053						
	Proportional and predictive reasoning	0.664	11	8.392	0.382	11	8.150	0.284	7	8.943	0.053						
	Connection to physics' concepts and principles	0.692	11	8.361	0.390	11	8.150	0.284	7	8.943	0.053						
	Real-world applicability and context	0.637	11	8.315	0.310	11	8.050	0.382	7	8.943	0.098						
	Relevancy	0.796	11	7.854	0.519	11	8.900	0.000	7	8.943	0.053						
Text - image - graph	Relevancy	0.627	5	8.140	0.371	5	8.520	0.432	4	8.700	0.476						
Text - image - table	Relevancy	0.729	5	8.280	0.536	5	8.540	0.451	4	8.275	0.914						
Text - table- graph	Relevancy	0.639	5	8.140	0.371	5	8.520	0.432	4	8.550	0.526						

levels), so students see the same symbols used in a uniform way. Taken together, these numbers suggest that the MPP are realistic, conceptually accurate, and appropriately scaled in difficulty, while keeping notation stable as a key quality for reliable physics questions.

The human evaluation results in Table 7 show strong performance across every MPP type. For image-text items, relevancy averaged 8.62 with a Cohen's κ of 0.83, and advanced questions nearly reached 8.99. Precision was also high at 8.61 ($\kappa = 0.66$); however, the easy level scored 8.20 because the questions at this level omit certain complicated or highly specialized information. By skipping these extra details, the problems become simpler and easier to read or solve. Text-graph tasks covered five sub-dimensions and stayed above 8.50 on every metric, with κ values between 0.65 and 0.82. Advanced graphs again dominated, posting 8.99 for accuracy versus 8.04 at the easy level, showing the system can add nuance without confusing labels. In text-table questions, clear labelling earned 8.49 ($\kappa = 0.80$), and advanced tables peaked at 8.94.

Easy and intermediate levels were slightly lower on reasoning and context because these levels are made simpler; they naturally scored a little lower on measures that reward deeper reasoning and richer context. Relevancy within the table set remained high at 8.85 overall, even though the easy level was 7.85. The mixed text-image-graph format recorded an 8.43 relevancy mean ($\kappa = 0.63$) and rose to 8.70 for advanced prompts. Questions that combine text, pictures, and tables got an average score of 8.31. The medium-level versions scored a little higher than the easy and hard ones because they showed just the right amount of visual information. Text-table-graph items came in at 8.39, climbing from 8.14 at the easy level to 8.55 at the advanced level. Across all categories, κ values from 0.63 to 0.83 indicate good agreement among the ten teacher raters. Advanced-level problems consistently score the highest, proving the generator scales complexity without losing clarity or correctness. Easy problems still sit comfortably above eight in most dimensions, so beginners receive clear, relevant, and fluent questions. Together these findings confirm that the GPT-4o system can reliably generate multimodal physics problems that support assessment, scaffolding, and personalized instruction for learners at different levels.

Correlation Analysis Between Automatic Evaluation and Human Evaluation

The result of Pearson correlation analysis of automatic and human evaluations is shown in Table 8. Relevancy measured by human evaluation shows a significant positive correlation with automatic evaluation metrics, especially with LSTM ($r = 0.628^{**}$) and BertF1 ($r = 0.358^{**}$). This suggests that the automatic methods, particularly those using sequential models, effectively capture semantic relevance consistent with human judgements. Diversity also shows a positive, though weaker, association with relevancy (e.g., BertF1 $r = 0.155$), indicating that more varied question generation tends to enhance relevancy. Complexity demonstrates a significant negative correlation with BertR ($r = -0.246^{**}$) and BertP ($r = -0.322^{**}$), implying that simpler sentence structures achieve higher semantic recall and precision. This is consistent with the idea that less complex language aids clear and effective communication. Complexity,

Table 8. Pearson Correlation Analysis Between Automatic Evaluation and Human Evaluation

Human/Teacher evaluation	Dimension	Sub-dimension	Automatic evaluation						
			A_BertFl	A_BertR	A_BertP	A_LSTM	A_Diversity		
Linguistic		T_Relevancy	0.358**	0.222**	0.1	0.628**	0.155		
		T_Complexity	-0.017	-0.246**	-0.322**	-0.005	-0.01		
		T_Diversity	0.266**	0.182*	0.063	0.403**	0.16		
		T_Completeness	0.166*	0.059	0.126	0.14	0.119		
		T_Fluency	0.131	0.059	0.058	0.698	0.088		
Physic aspect		T_Answerability_Physics	0.017	0.019	0.031	0.712**	0.058		
		T_Difficulty_Level_Physics	-0.093	-0.200*	-0.174*	-0.039	-0.005		
		T_Realistic_Physics	0.098	0.036	0.05	0.703**	0.08		
		T_Concept_Understanding	0.491**	0.308**	0.138	0.586**	0.181*		
		T_Variable_Name_Consistency	0.213**	0.116	0.076	0.681**	0.11		
Text - graph question		T_Graph_Accuracy_and_Proper_labeling	0.333*	0.223	0.205	0.196	0.326*		
		T_Graph_Sense_in_Physical_contexts	0.246	0.161	0.192	0.171	0.371*		
		T_Scale_and_Proportions	0.233	0.15	0.188	0.163	0.370*		
		T_Creativity_and_Critical Thinking	0.216	0.136	0.189	0.143	0.373*		
		T_Axis_and_Text_Relevancy	0.364*	0.246	0.2	0.17	0.306*		
Text- table question		T_Clear_labels_and_scales	0.498**	0.183	0.269	0.24	0.234		
		T_Physics_Connection	0.344	0.15	0.163	0.03	0.151		
		T_Real_World_Applicability	0.314	0.129	0.165	0.033	0.208		
		T_Proportional_and_predictive_reasoning	0.301	0.074	0.208	0.127	0.309		
		T_Column_and_Text_Relevancy	0.506**	0.204	0.257	0.224	0.259		
Image - text question		T_Precision_Image	0.332	0.226	0.118	0.406*	0.308		
		T_Relevancy_Image	0.412*	0.25	0.203	0.301	0.431**		
	Ablation	Text_Image_Table_Relevancy	0.336	0.350	0.158	0.236	0.659*		
		Text_Image_Graph_Relevancy	0.666**	0.691**	0.491	-0.180	0.461		
		Text_Graph_Table_Relevancy	0.484	0.405	0.479	0.379	0.304		

Correlations marked ** are significant at p < .01; those with * are significant at p < .05.

however, shows no meaningful correlation with LSTM or diversity scores, suggesting these models may be less sensitive to sentence level of complexity.

Furthermore, human evaluations of diversity are positively related to automatic diversity metrics ($r = 0.16$) and LSTM ($r = 0.403^{**}$), which means the automatic system can recognize differences in generated problems that match how humans see diversity. Completeness has a small positive relationship with BertF1 ($r = 0.166^*$), but not as much with other measures, which means that automatic evaluations somewhat reflect how well questions provide the necessary information. The difficulty level in physics exhibits a significant negative correlation with BertR ($r = -0.200^*$) and BertP ($r = -0.174^*$), suggesting that simpler problems tend to achieve better recall and precision in automatic evaluation. Concept understanding is strongly linked to both BertF1 and LSTM, showing that these models do a good job of recognizing the depth of understanding that human evaluators see. Automatic diversity metrics consistently showed moderate positive relationships with all graph features, meaning they are good at measuring how varied and relevant the questions are.

Additionally, A-BertF1 demonstrated strong correlations with accuracy and axis relevancy, reflecting its ability to assess clear labelling and meaningful graph content. A_BertF1 showed strong positive correlations with clear labels, scales, and column-text relevancy in table-based questions, indicating it effectively evaluates clarity and relevance. A_BertF1 showed a significant positive correlation with image relevancy (0.412^*). A_LSTM showed a significant positive correlation with image precision (0.406^*), indicating its effectiveness in assessing detailed image quality. Meanwhile, A_Diversity had the strongest correlation with image relevancy (0.431^{**}), suggesting the diversity of the questions are diverse, not similar. A_BertF1 and A_BertR showed strong positive correlations with relevancy in the Text-Image-Graph modality (0.666^{**} and 0.691^{**}), indicating high accuracy in evaluating this combination modality. Additionally, A_Diversity had a significant positive correlation with text-image-table relevancy (0.659^*), highlighting its role in capturing variety and relevance.

In summary, the automatic evaluation metrics show meaningful and significant correlations with human judgements across multiple linguistic and physical dimensions. These findings show that the automatic evaluation system is a trustworthy substitute for human evaluation, accurately representing problem relevance, understanding of concepts, and diversity, while also pointing out that as complexity increases, the difficulty level tends to decrease. This confirms the system's suitability for assessing and guiding the generation of high-quality multimodal physics word problems.

Discussion and Implications

The findings provide strong evidence that the P-MAGIC system can generate high-quality Multimodal Physics Problems (MPPs) across different difficulty levels. For Research Question 1, the automatic evaluation results indicate consistently strong semantic alignment and linguistic quality, as reflected in high overall BERT F1, recall, and precision scores, together with acceptable diversity and LSTM-based fluency scores. Importantly, these scores remain relatively stable across easy, intermediate, and

advanced levels, suggesting that the system maintains semantic relevance even when cognitive demand changes. This stability is important because it shows that the instructional prompt design and difficulty-control mechanism are functioning as intended, rather than producing good outputs only at one level of complexity. At the same time, the ablation results add a deeper insight: image-based modalities (especially text-image and text-image-graph combinations) perform better than some graph- or table-dominant combinations, and annotated modalities consistently outperform non-annotated ones. This suggests that multimodality alone is not enough; the pedagogical value depends on how clearly the representations are designed and annotated.

For Research Question 2, the human evaluation reinforces and extends the automatic findings by showing that teachers judged the generated MPPs as pedagogically strong across linguistic, physics, and multimodal dimensions. The scores reported for image-text, graph-text, table-text, and mixed modalities are generally high (mostly above 8 on a 10-point scale), while Cohen's kappa values indicate good inter-rater agreement, which strengthens the credibility of the teacher ratings. A meaningful pattern also appears across difficulty levels: advanced-level items tend to receive the highest ratings in several dimensions, while easy-level items score slightly lower on aspects such as precision, reasoning, or contextual richness. This should not be interpreted as a weakness of the system, but rather as evidence that the generator is actually respecting the intended instructional simplification at lower levels. In other words, the easier items appear less dense because they are designed to reduce cognitive load for beginners, while advanced items are richer in representational nuance and reasoning demand. This supports the educational usefulness of P-MAGIC as a scaffolded question-generation tool rather than a one-size-fits-all generator.

For Research Question 3, the correlation analysis shows that automatic and human evaluations are aligned in meaningful—but not uniform—ways, which is exactly what we would expect in a valid hybrid evaluation framework. The strongest pattern is that LSTM-based automatic scores correlate substantially with teacher judgments on key pedagogical qualities such as relevancy, answerability, realism, concept understanding, and variable-name consistency. BERT-based metrics also correlate with several human dimensions, especially concept understanding and multimodal relevancy in text-image-graph formats. However, the correlations are not equally strong for every sub-dimension, and some difficulty/complexity indicators show weak or negative correlations with certain BERT metrics. This is an important discussion point: automatic metrics are very useful for screening relevance, semantic alignment, and general quality at scale, but they are less reliable for judging nuanced pedagogical constructs such as cognitive complexity or appropriateness of difficulty without human interpretation. Therefore, the results do not suggest replacing teachers; rather, they support a layered evaluation workflow in which automatic metrics handle rapid large-scale filtering and teachers focus on deeper pedagogical validation.

These findings have several implications for practice and future research. For classroom practice, the results suggest that teachers should prioritize multimodal problem design that includes images and explicit annotations, because these features appear to improve clarity, semantic relevance, and likely student engagement. For

system development, the findings justify integrating automatic metrics (especially LSTM- and BERT-based measures) as a real-time quality-control layer in MPP generation pipelines, while still retaining teacher review for final deployment in high-stakes learning contexts. For research, the study supports a broader claim that multimodal, context-aware AI systems can move beyond generic question generation toward pedagogically grounded assessment support. At the same time, the current evidence is still limited to centripetal acceleration and expert evaluation settings; therefore, future studies should test the system in classroom interventions, examine student learning gains, and investigate whether the same evaluation patterns hold for other physics topics and other multimodal representations. In this sense, P-MAGIC should be viewed not only as a question generator, but as a platform for studying how AI, multimodal representation, and teacher expertise can be combined to improve physics learning design.

Conclusion and Future Work

Overall, the study shows that P-MAGIC app generated MPPs based on authentic contextual recognition performs well in both automatic and human evaluations. The automatic evaluation results indicate that the system produces questions with high relevancy, diversity, and excellent sentence quality across different difficulty levels. Human evaluations by experienced physics teachers confirm that the generated problems maintain strong language quality, physical aspects, and validity across various question types involving images, graphs, and tables. Furthermore, the significant positive correlations found between automatic and human evaluations suggest that automatic metrics can reliably reflect human judgement in assessing MPP quality.

These results address the research questions by showing substantial alignment between automatic and human evaluation outcomes, supporting the feasibility of using P-MAGIC to generate useful and varied physics problems tailored to different learning levels. Nevertheless, the study is subject to limitations, particularly the restricted scope of physics topics considered, which may constrain the generalizability of the findings. Future work should expand the coverage to a broader range of physics concepts and investigate classroom deployment to further validate the system under authentic instructional conditions. In addition, future research should conduct comparative analyses and implement quasi-experimental designs with senior high school students to examine the extent to which P-MAGIC influences physics learning performance. Finally, further opportunities include involving students more directly in the MPP generation process and leveraging hybrid evaluation workflows that combine human expertise with language-model assistance to enhance both the scalability and robustness of the evaluation process.

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Ethical Considerations

Research ethics approval was obtained by the Institutional Review Board of National Central University. This study was conducted in accordance with the Academic Integrity Code of National Central University.

Consent to Participate

Informed consent statements were collected from all teachers who participated in this study.

Authors' Contribution

WYH and VKRS conceptualized the study; VKRS developed the backend system, conducted experiments, performed data analysis, and wrote the manuscript; WYH and SWDP reviewed and edited the manuscript; MWA contributed to the front-end UI development. All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Supplemental Material

Supplemental Material for this article is available online.

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